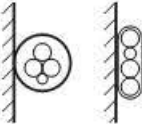
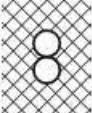
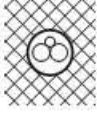
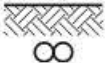


TABLE 3.1
CABLE TYPES AND THEIR APPLICATION IN WIRING SYSTEMS

Installation method		Description		Typical cable types
		Unenclosed	On a surface (including cable tray or ladder)	Insulated and sheathed Screened or armoured
				
			On a surface partly surrounded by thermal insulation	Mineral insulated, metal sheathed (MIMS) Earthing conductors
			Fully surrounded by thermal insulation	
			Buried direct in the ground, subject to the requirements of Clause 3.11	Insulated and sheathed, screened or armoured, earthing conductors

(continued)

TABLE 3.1 (continued)

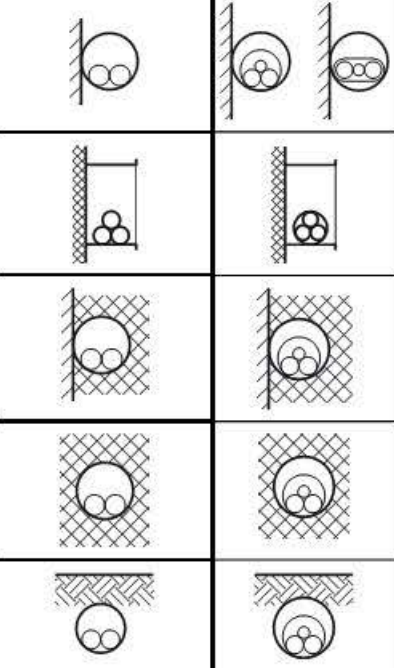
Installation method		Description		Typical cable types
			On a surface (including cable trunking)	Insulated, unsheathed Insulated and sheathed Screened or armoured MIMS Earthing conductors
				
				
				
			Underground, subject to the requirements of Clause 3.11	Insulated, unsheathed Insulated and sheathed Screened or armoured Earthing conductors
		Supported on a catenary system		Insulated and sheathed Screened or armoured Earthing conductors
		Supported on insulators		Aerial conductors Earthing conductors

TABLE 3.2
LIMITING TEMPERATURES FOR INSULATED CABLES

Type of cable insulation ⁽¹⁾	Operating temperature of conductor, °C		
	Normal use ⁽²⁾	Maximum permissible ⁽⁷⁾	Minimum ambient ⁽³⁾
Thermoplastic ⁽⁴⁾			
V-75	75	75	0
HFI-75-TP, TPE-75	75	75	-20
V-90	75	90	0
HFI-90-TP, TP-90	75	90	-20
V-90HT	75	105	0
Elastomeric			
R-EP-90	90	90	-40
R-CPE-90, R-HF-90, R-CSP-90	90	90	-20
R-HF-110, R-E-110	110	110	*
R-S-150	150	150	-50
Cross-linked polyethylene			
X-90, X-90UV, X-HF-90	90	90	*
X-HF-110	110	110	*
MIMS ⁽⁵⁾	100	250	(6)
Other types			
PE, LLDPE	70	70	*

* Refer to manufacturer's information.

NOTES:

- 1 The types of cable insulation given in Table 3.2 are included in relevant specifications, i.e. the AS/NZS 5000 series, AS/NZS 3191, AS/NZS 3808 and AS/NZS 60702.1.
- 2 Lower maximum temperatures will apply where materials used in the construction of the cables or in association therewith, such as coverings, sheathings, insulating sleeving on connections and sealing compounds, have maximum operating temperatures lower than the cable proper. However, the allowable operating temperatures for such materials shall not be exceeded.
- 3 If manufacturer's recommendations permit, cables may be installed in locations where temperatures lower than specified may occur.
- 4 The normal operating temperature of thermoplastic cables, including flexible cords installed as installation wiring, is based on a conductor temperature of 75°C. This is because of the risk of thermal deformation of insulation if the cables are clipped, fixed or otherwise installed in a manner that exposes cable to severe mechanical pressure at higher temperatures.

TABLE 3.3
NOMINAL MINIMUM CROSS-SECTIONAL AREA OF CONDUCTORS

Type of wiring system	Use of the circuit	Conductor	
		Material	Cross-sectional area mm ²
Insulated conductors	Socket-outlets (see <i>Exception 1</i>)	Copper	2.5
	Other circuits		1
	Signal and relay control circuits		0.5
Bare conductors	—	Copper	6
Insulated flexible conductors	—	Copper	0.75
Aerial wiring	—	Copper	6
		Aluminium	16

Exceptions:

- 1 *Smaller conductors may be used on subcircuits supplying socket-outlets, based on their suitability, in accordance with this Standard, and taking account of voltage drop, current-carrying capacity and reliability of connections.*
- 2 *Table 3.3 does not limit cable sizes for extra-low voltage or switchboard wiring.*

NOTE: The size of unprotected consumer mains should be coordinated with the electricity distributor.

3.5.2 Neutral conductor

The minimum size of the neutral conductor shall be as follows:

- (a) *Single-phase two-wire circuit* The neutral conductor or conductors of a single-phase consumer main, submain or final subcircuit shall have a current-carrying capacity not less than—
 - (i) the current-carrying capacity of the associated active conductor; or
 - (ii) the total current to be carried, where there is more than one active conductor.
- (b) *Multiphase circuit* The current-carrying capacity of the neutral conductor of a multiphase circuit shall not be less than that determined in accordance with the following:
 - (i) *Harmonic currents* Where a consumer main, submain or final subcircuit supplies a substantial load that generates harmonic currents, e.g. fluorescent lighting, computers, soft starters, variable speed devices or other electronic devices, the third and any higher order harmonic current generated in the equipment

Conductors with green, yellow or green/yellow combination coloured insulation or sheathing shall not be used as active or neutral conductors in installation wiring. In New Zealand, use of these colours is restricted for conductors but not for sheathing. **NZ**

Exception: The colour identification provisions of Table 3.4 need not apply to the special applications listed in Clause 3.8.3.

- * In New Zealand, there is no restriction on sheathing colour. **NZ**

NOTES:

- 1 Internal wiring of equipment is not regarded as installation wiring but may be subject to particular equipment standards.
- * 2 Switchboard wiring is not regarded as installation wiring but the AS/NZS 3439 series and AS/NZS 61439 series restrict the green/yellow combination to the identification of earthing conductors.

TABLE 3.4
CONDUCTOR COLOURS FOR INSTALLATION WIRING

Function	Insulation colour
Protective earth	Green/yellow
Equipotential bonding	Green/yellow
Neutral	Black or light blue
Active	Any colour other than green, yellow, green/yellow, black or light blue

NOTES:

- 1 When green/yellow is used, one colour shall cover not less than 30% and not more than 70% of the surface area, with the other colour covering the remainder of the surface.
- 2 Recommended colours for actives are—
 - (a) Red or brown for single-phase; or
 - (b) Red, white or blue for multiphase.
- 3 Where colours are used for the identification of cable cores, Australian and New Zealand cable identification colours and European cable identification colours shall not be combined within the same wiring enclosure or the same multi-core cable.
- * 4 In New Zealand domestic installations, the only permitted colour for neutral conductors is black. **NZ**

3.8.2 Colour identification

3.8.2.1 Colour identification by sleeving or other means

Colour identification by sleeving or other means, using colours corresponding to those listed in Table 3.4 at each termination, may be used as a means of identification for the following purposes:

- (a) Conductors with black or light blue insulation used as active conductors.

TABLE 3.5
UNDERGROUND WIRING SYSTEM CATEGORIES

1	2	3	4	5	6	7	8	9
Method of protection— Type of cable	Heavy-duty conduit*	Medium-duty conduit* encased in concrete	Heavy-duty fibre cement conduit	Fibre cement conduit encased in concrete	Medium- or heavy-duty galvanized pipe	Medium-duty, corrugated or flexible conduit*	Buried direct in the ground with no enclosure	Chased in rock with no enclosure
Insulated, unsheathed conductors	A	A	NP	NP	NP	B	NP	NP
Insulated and sheathed conductors	A	A	A	A	A	B	B	NP
Sheathed, armoured and served cables	A	A	A	A	A	A	A	C
Neutral-screened cables suitable for underground	A	A	A	A	A	A	A	NP
Neutral-screened cables	A	A	A	A	A	B	B	NP
Served MIMS cables	A	A	A	A	A	B	B	C
Aluminium sheathed or strip armoured cables with PVC sheath	A	A	A	A	A	B	B	C

* These conduits and any associated fittings consist of insulating material only and do not have any conductive components.

KEY:

A = Category A wiring system

B = Category B wiring system

C = Category C wiring system

NP = Not permitted

3.11.4 Installation requirements

3.11.4.1 General

Underground wiring systems shall be installed in accordance with the requirements of Clauses 3.11.4.2 to 3.11.4.6.

- * NOTE: Typical arrangements of underground wiring systems are shown, and measurement values also apply to any direction where the cover is relied upon to minimize risk of excavation.

3.11.4.2 Unenclosed cables

Any Category A or Category B wiring system that comprises cables not installed in a wiring enclosure shall be laid on a bed of not less than 50 mm of sand or friable soil, free of sharp stone, and covered by not less than 50 mm of the same material.

3.11.4.3 Category B wiring system mechanical protection

For a Category B wiring system, additional mechanical protection shall be provided as follows:

- (a) The protection shall be placed not more than 75 mm above the wiring system.
- (b) The protection shall be not less than 150 mm wide.
- (c) The protection shall overlap the wiring system by at least 40 mm on each side.
- (d) The protection shall consist of one or a combination of the following:
 - (i) Precast concrete slabs having a thickness of not less than 40 mm and a classification of not less than grade 20 in accordance with AS 3600 or NZS 3104.
 - (ii) Concrete slabs cast on-site having a thickness of not less than 100 mm.
 - (iii) A continuous concrete pour having a thickness of not less than 75 mm.
 - (iv) Fibrous cement slabs having a thickness of not less than 12 mm.
 - (v) Bricks manufactured specifically for the protection of electric cables.
 - (vi) Polymeric cable cover strips complying with AS 4702.
 - (vii) Other materials that offer the same degree of protection afforded by the materials in Items (i) to (vi).

3.11.4.4 Minimum depth of cover

Underground wiring systems shall be installed with the minimum depth of cover and protection specified in Table 3.6.

These dimensions shall apply vertically between the upper surface of—

- (a) the wiring system for a Category A or Category C system; or
- (b) the additional mechanical protection of a Category B system,

and the surface of the ground or below any poured concrete laid on that surface (see Figures 3.10 to 3.17).

- * Where cables are buried close to a sloping or vertical surface, these dimensions shall also apply perpendicular to that surface (see Figure 3.17).

TABLE 3.6

UNDERGROUND WIRING SYSTEMS—MINIMUM DEPTH OF COVER

Location of wiring system	Covering on surface of ground above wiring system	Cat A system	Cat B system	Cat C system
Within confines of a building	Poured concrete of 75 mm minimum thickness	0 mm (directly below)	0 mm (directly below)	0 mm (directly below)
	No surface covering or less than 75 mm thickness of concrete	500 mm	500 mm	50 mm
Elsewhere external to a building	Poured concrete of 75 mm minimum thickness	300 mm	300 mm	50 mm
	No surface covering or less than 75 mm thickness of concrete	500 mm	500 mm	50 mm

3.11.4.5 Identification of underground wiring

Wiring systems installed underground shall be identified by an orange marker tape complying with AS/NZS 2648.1. In order to provide early detection of the presence of underground wiring during excavation work, marker tape shall be positioned at approximately 50% of the depth of cover above the wiring system or any additional mechanical protection provided for that system.

Where the wiring system is chased in rock, orange marker tape shall be laid directly on top of the wiring system before the concrete is poured.

Exception: Marker tape may be omitted where an underground wiring system is installed by boring provided that—

- (a) *the location of the wiring system is marked and recorded in a suitable permanent location that is readily available to any person involved with excavation work at the location of the wiring system; and*
- (b) *if an enclosure is retained after boring operations, the enclosure is coloured orange.*

3.11.5 Spacing from other underground services

All underground wiring systems shall be spaced not less than 100 mm from other underground services.

Wiring systems shall be suitably marked with warning tape in accordance with Clause 3.11.4.5 and maintain a separation from telecommunications, gas and water services in accordance with Table 3.7.

Further requirements for the separation of telecommunications, gas and water systems from low voltage wiring systems are provided in Clause 3.9.8.4.

TABLE 3.7
MINIMUM SEPARATION OF UNDERGROUND SERVICES

Type of service	Minimum separation to low voltage electrical service identified in accordance with Clause 3.11.4.5 and mechanically protected ⁽¹⁾	Minimum separation of conductive enclosures to low voltage electrical earthing electrode ⁽¹⁾
	mm	mm
Water service not greater than DN65 ⁽²⁾	100	500
Water service greater than DN65 ⁽²⁾	300	500
Sanitary drainage	100	500
Stormwater drainage	100	600
Gas	100	500
Telecommunications	100	

NOTES:

- 1 Authorities, such as water and gas suppliers and electricity distributors, may require their services to be spaced at a greater distance from underground wiring systems. Also see Figure 3.10.
- 2 DN = internal diameter of pipe.

Exceptions:

- 1 *Two or more underground wiring systems may be grouped together where they are associated with the same electrical installation.*
- 2 *The requirements of this Clause (Clause 3.11.5) may be varied where a number of services are installed touching in a common trench, provided that each service is installed in a separate enclosure that identifies the service.*
- 3 *Separation distances between conductive enclosures and the earthing electrode are not required where all conductive enclosures are bonded within the installation.*

TABLE 3.8
MINIMUM AERIAL CONDUCTOR CLEARANCES

All dimensions in metres

Type of aerial conductor	Minimum height above buildings, structures, ground or elevated areas				From buildings—Horizontal clearance from walls, etc.	From clothes lines, radio and television aerials, counter-poise or stay wires	From tele-communications lines ⁽²⁾	Above swimming pools	Above areas where sailing craft, or irrigation pipes are used ⁽³⁾
	Over areas used by vehicles	Over areas not used by vehicles	Over roofs used for traffic or resort	Over other roofs and structures					
Bare live conductors	5.5	5.0	3.7	3.0	2.0	2.0	1.2	Not permitted	Not permitted
Insulated and unsheathed live conductors	4.6	3.0	3.0	2.0	1.0	2.0	0.6	3.0	5.5
Neutral-screened cable	4.6	3.0	2.7	0.5	1.0	2.0	0.6	3.0	4.5

NOTES:

- 1 When erecting aerial conductors an allowance for sag and sway under operating conditions needs to be added to ensure that the above clearances are maintained. (Refer Clause 3.12.3 and Appendix D.)
- 2 Further information regarding required clearances for crossing telecommunication lines is contained in AS/CA S009.
- 3 Warning notices shall be erected where required by Clause 3.12.3.2.
- 4 Increased distances may be required over public roadways.

3.12.4 Distance between supports (spans)

The length of span of aerial conductors shall not exceed the values specified in Table 3.9 for the appropriate type and size of conductor.

Exception: Spans greater than the values specified in Table 3.9 may be used, provided that the design is in accordance with sound engineering practice.

NOTES:

- 1 An indication of acceptable stringing practice is given in Table D2 of Appendix D.
- 2 More detailed information is available in AS/NZS 7000.

TABLE 3.9
AERIAL CONDUCTOR MAXIMUM SPANS

Type of conductor	Size mm ²	Maximum span m
Insulated annealed copper including neutral-screened	≥6	20
Bare hard-drawn copper	≥6	60
Insulated hard-drawn copper including two-, three- and four-core twisted but excluding neutral-screened	6	40
	10	50
	≥16	60
Neutral-screened cables with hard-drawn copper conductors—		
two conductors	6 or 10	40
three conductors	6 or 10	60
four conductors	6 or 10	50
two, three, or four conductors	16	60
Insulated or bare aluminium excluding neutral-screened	16	50
	≥25	60
Aerial bundled cables (aluminium conductor)	≥25	60

3.12.5 Aerial conductor supports

3.12.5.1 General

Supports for aerial conductors shall be insulators or purpose-designed fittings suitable for the type of cable with which they are used.

3.12.5.2 Pin-type insulators

Pin-type insulators shall not be used for supporting aerial conductors where—

- (a) the strain tends to lift or otherwise separate the conductors from the insulators; or

- (b) the direction of the conductors is changed by more than 30°.

3.12.5.3 Hardware

Any hardware or fittings used in association with the aerial line shall be of corrosion-resistant material, or other material suitably protected against corrosion.

3.12.5.4 Spacing between conductors

Conductors shall be adequately spaced to prevent contact with each other under all conditions of sag and sway.

The spacing between conductors at supports, measured in any direction, shall be not less than that shown in Table 3.10.

Exception: The spacing between conductors of a multi-core cable or cables operated in parallel may be less than that shown in Table 3.10.

NOTE: The electricity distributor may require a minimum clearance between consumer aerial lines and any electricity distributor aerial lines.

TABLE 3.10
SPACING BETWEEN AERIAL
CONDUCTORS AT SUPPORTS

Span	Spacing m	
	Insulated conductors	Bare conductors
≤10	0.2	0.4
>10 ≤25	0.3	0.5
>25 ≤45	0.4	0.6
>45 ≤60	0.5	0.7

3.12.6 Poles and posts (including supports, struts and extensions to structures)

Poles and posts shall be constructed of materials suitable for the conditions of use, taking account of the following:

- (a) Size.
- (b) Depth in ground.

NOTES:

- 1 Guidance on the size of a typical range of poles and posts is given in Appendix D.
- 2 Guidance on the depth in ground for a typical range of poles and posts of lengths up to 7 m above the ground is given in Appendix D.

The depth in ground may be reduced if the pole or post is set in solid rock, provided that the arrangement is not inferior to installation in accordance with the above requirements.

NOTES:

- 1 Guidance on materials suitable for restoring fire-rated constructions is given in national building codes.
- 2 National building codes may have requirements for restoring acoustic insulation that has been penetrated by electrical equipment.

4.2.2.7 Thermal insulation—New Zealand only**NZ**

In New Zealand only, appliances and accessories in domestic installations shall be installed on the assumption that thermal insulation in ceilings, walls and under floors, if not currently installed, will be installed in the future.

4.2.3 Protection against burns

An accessible part of electrical equipment within arm's reach shall not attain a temperature in excess of the appropriate limit stated in Table 4.1.

Each accessible part of the electrical installation that may, even for a short period, attain a temperature exceeding the appropriate limit in Table 4.1 under normal load conditions shall be guarded so as to prevent accidental contact.

Exceptions:

- 1 *This requirement need not apply to electrical equipment that complies with a limiting temperature specified in an appropriate Standard.*
- 2 *This requirement does not apply to items such as lamps.*

TABLE 4.1
TEMPERATURE LIMITS IN NORMAL SERVICE
FOR PARTS OF ELECTRICAL EQUIPMENT WITHIN ARM'S REACH

Accessible part	Material of accessible surface	Maximum temperature °C
Hand-held means of operation	Metallic	55
	Non-metallic	65
Parts intended to be touched but not hand-held	Metallic	70
	Non-metallic	80
Parts that need not be touched for normal operation	Metallic	80
	Non-metallic	90

4.3 CONNECTION OF ELECTRICAL EQUIPMENT**4.3.1 General**

Electrical equipment may be connected to the installation wiring by one of the methods detailed in Clauses 4.3.2 to 4.3.5.

TABLE 4.3

* GUIDE TO RECESSED LUMINAIRE CLASSIFICATION, APPLICATION AND GENERAL RESTRICTIONS ON USE

Type	Abutted ⁽¹⁾	Normal use ⁽²⁾ —covered	Use with insulation ⁽³⁾	Accessibility to high temperature parts ⁽⁴⁾	Surface of luminaire normal operating temperature limit	Comments
Non-IC	No	No	No	Fully accessible	No limit on side or top surface of luminaire Mounting surface of luminaire limited to 90°C	Not for residential use, intended for non-residential use only Non-IC luminaires are prohibited in residential dwellings or other places where building insulation may be installed (now or in the future) These luminaires have not been tested to show they are safe for use with building insulation
Do-not-cover	No	No	Yes	Fully accessible	No limit on side or top surface of luminaire Mounting surface of luminaire limited to 90°C	Suitable for residential or non-residential use in Australia Prohibited from installation in residential locations in New Zealand Cannot be covered Clearance distances from sides of luminaire to insulation and clearance above luminaire are to be in installation instructions supplied with the luminaire and have to be adhered to for correct installation

(continued)

TABLE 4.3 (continued)

Type	Abutted ⁽¹⁾	Normal use ⁽²⁾ —covered	Use with insulation ⁽³⁾	Accessibility to high temperature parts ⁽⁴⁾	Surface of luminaire normal operating temperature limit	Comments
						These luminaires have been tested to show they are safe for use with building insulation present (observing manufacturer's stated installation clearance distances) They are not to be covered in building insulation; however, have been tested to show if inadvertently covered they should not become a fire hazard
CA90	Yes	No	Yes	Limited access In this Standard, this is assessed for access to high temperature parts by use of a 5.6 mm probe to side and top of luminaire	90°C limit on side or top or mounting surface of luminaire	Suitable for residential or non-residential use in Australia and New Zealand Cannot be covered Insulation can be placed against the sides of the luminaire Manufacturer's clearance above luminaire to be observed for correct installation These luminaires have been tested to show they are safe for use with building insulation present and placed against the sides of the luminaire They are not to be covered in building insulation; however, have been tested to show if inadvertently covered they should not become a fire hazard

(continued)

TABLE 4.3 (continued)

Type	Abutted ⁽¹⁾	Normal use ⁽²⁾ —covered	Use with insulation ⁽³⁾	Accessibility to high temperature parts ⁽⁴⁾	Surface of luminaire normal operating temperature limit	Comments
CA135	Yes	No	Yes	Some access In this Standard, this is assessed for access to high temperature parts by use of a 50 mm probe to side and top of luminaire	135°C limit on side or top surface of luminaire Mounting surface of luminaire limited to 90°C	Not permitted for use in Australia New Zealand only—suitable for residential or non-residential use Cannot be covered Insulation can be placed against the sides of the luminaire Manufacturer's clearance above luminaire to be observed Cannot be used where building materials or insulation is not rated for exposure to constant temperatures of 135°C (cannot have such materials that may be placed so as to be touching the sides of the luminaire) These luminaires have been tested to show they are safe for use with building insulation present and placed against the sides of the luminaire They are not to be covered in building insulation; however, have been tested to show if inadvertently covered they should not become a fire hazard

(continued)

TABLE 4.3 (continued)

Type	Abutted ⁽¹⁾	Normal use ⁽²⁾ —covered	Use with insulation ⁽³⁾	Accessibility to high temperature parts ⁽⁴⁾	Surface of luminaire normal operating temperature limit	Comments
IC	Yes	Yes	Yes	Limited access In this Standard, this is assessed for access to high temperature parts by use of a 5.6 mm probe to sides and top of luminaire	90°C limit on side or top or mounting surface of luminaire	Suitable for residential or non-residential use in Australia and New Zealand Used where some air transfer is allowed or desired between living space and roof space (there will be some air transfer between the spaces if the luminaire is not fully covered in insulation) These luminaires have been tested to show they are safe in normal use when covered in building insulation and abutted to combustible building elements
IC-4	Yes	Yes	Yes	Restricted access In this Standard, this is assessed for access to high temperature parts by use of a IP4X—1 mm probe to sides and top and front face of luminaire	90°C limit on side or top or mounting surface of luminaire	Suitable for residential or non-residential use in Australia and New Zealand Used where air transfer is not permitted or not desired between living space and roof space (there will be no air transfer between spaces even if there is no insulation covering the luminaire) Typical use is passive house design where no air transfer is allowed These luminaires have been tested to show they are safe in normal use when covered in building insulation and abutted to combustible building elements

(continued)

TABLE 4.3 *(continued)*

Type	Abutted ⁽¹⁾	Normal use ⁽²⁾ —covered	Use with insulation ⁽³⁾	Accessibility to high temperature parts ⁽⁴⁾	Surface of luminaire normal operating temperature limit	Comments
No marking	No	No	No	No		Not verified as tested/compliant to Australian/New Zealand Standards Marking is required by standard—no marking indicates non-compliance Do not install any luminaire that does not have one of the marking symbols or instructions specifying any clearance distances

NOTES:

- 1 May be abutted against normally flammable building elements or insulation.
- 2 Intended and tested to be used under building insulation as part of normal operation.
- 3 May be used where building insulation may be installed (now or in the future).
- 4 Classification and probe to determine access of insulation, etc. to high temperature parts.

K = factor dependent on the material of the protective earthing conductor, the insulation and other parts, and the initial and final temperatures

NOTES:

- 1 Values of K for protective earthing conductors in various conditions of service are given in the AS/NZS 3008.1 series, e.g. for copper conductors not laid up with other conductors, with PVC insulation $K = 136$, or for bare copper conductors $K = 170$.
- 2 Examples of the application of this equation are contained in the AS/NZS 3008.1 series.

If application of the equation produces non-standard sizes, conductors of the nearest higher standard cross-sectional area shall be used.

NOTE: Maximum permissible temperatures for joints should be considered (see the AS/NZS 3008.1 series).

TABLE 5.1
MINIMUM COPPER EARTHING CONDUCTOR SIZE

Nominal size of active conductor mm ²	Nominal size of copper earthing conductor, mm ²	
	With copper active conductors	With aluminium active conductors
1	1*	—
1.5	1.5*	—
2.5	2.5	—
4	2.5	—
6	2.5	—
10	4	—
16	6	4
25	6	6
35	10	6
50	16	10
70	25	10
95	25	16
120	35	25
150	50	25
185	70	35
240	95	50
300	120	70
400	≥120†	≥95†
500	≥120†	≥95†
630	≥120†	≥120†
>630	≥25% of active size†	≥25% of active size†

* These earthing conductors shall only be used where incorporated in a multi-core cable or flexible cord, other than a lift travelling cable, in accordance with Clause 5.3.3.4, Items (b) and (c).

† A larger earthing conductor may be required to satisfy Clause 5.3.3.1.1.

5.3.3.2 Main earthing conductor

Where the main earthing conductor connects the main earth terminal to an electrode (arrangement depicted in Figure 5.1), the conductor shall be determined from Table 5.1 in relation to the cross-sectional area of the largest active conductor of the consumer mains.

The cross-sectional area of such a copper main earthing conductor shall be not less than 4 mm² and need not be greater than 120 mm².

Exceptions: The size of the main earthing conductor need not be determined in relation to the size of the largest active conductor of the consumer mains under the following conditions:

- 1 *Where double insulation is maintained between the point of supply and the load terminals of the protective devices for the submains and final subcircuits outgoing from the main switchboard. The minimum size of the main earthing conductor may be determined in relation to the cross-sectional area of the largest active conductor of the largest outgoing submain or final subcircuit.*
- 2 *Where the cross-sectional area of the consumer mains is larger than that required to carry the maximum demand of the installation because of voltage-drop limitations. The minimum size of the main earthing conductor may be determined in relation to the cross-sectional area of the minimum cable size required to carry the maximum demand.*

Where the conductor connects the main earth bar to an earth bar within a substation (arrangement depicted in Figure 5.2), the conductor shall be determined from Table 5.1 or by calculation.

NOTE: In this application, the main earthing conductor forms part of the earth fault current path and its size needs to be determined based on the prospective fault current level and duration.

5.3.3.3 Aerial earthing conductors

The minimum size of a protective earthing conductor installed as an aerial conductor shall be selected in accordance with Clause 5.3.3.1. In addition, the cross-sectional area of the earthing conductor shall be not less than the size specified in Clause 3.12.2 and Table 3.9 according to the type of conductor and length of span.

5.3.3.4 Earthing conductors in cables, flexible cables or flexible cords

The minimum size of a protective earthing conductor in the form of a cable, flexible cable or flexible cord shall be selected in accordance with Clause 5.3.3.1.2 and, in addition, the cross-sectional area of the earthing conductor shall comply with the following requirements appropriate to the type of cable or cord:

NOTE: AS/NZS 3191 precludes the use of aluminium conductors in flexible cords.

- (a) *Single-core cable, flexible cable or flexible cord* The minimum size of a copper earthing conductor in the form of a single-core insulated cable, flexible cable or flexible cord shall be 2.5 mm².

Exception: Where connections in a multi-core cable require a short length of single-core insulated earthing conductor to be unenclosed, the earthing conductor size may be not less than 1 mm².

- (b) *Multi-core cable or flexible cable* The minimum size of an earthing conductor incorporated with associated live conductors in a sheathed multi-core cable or flexible cable shall be 1 mm².

- (c) *Multi-core flexible cord* The minimum size of an earthing conductor incorporated with the associated live conductors in the sheath of a multi-core flexible cord, shall be not less than the cross-sectional area of the largest active conductor incorporated in the flexible cord, provided that—

- (i) the cross-sectional area of the largest active conductor incorporated in the flexible cord is more than 0.5 mm² and less than 2.5 mm²; and
- (ii) the flexible cord is used to supply a hand-held or portable appliance.

- (d) *Screen and drain wires* The minimum cross-sectional area of any screen or drain wire incorporated in a sheathed cable, flexible cable or flexible cord shall be selected in accordance with Clause 5.3.3.1.2 and, in addition, shall be not less than that required by this Clause (Clause 5.3.3.4) for a protective earthing conductor.

Exception: This requirement need not apply where the screen or drain wire is separated from live conductors by double insulation or the equivalent thereof.

5.3.4 Main earthing terminal/connection or bar

In every electrical installation, a main earthing terminal/connection or bar shall be provided at the main switchboard. The following conductors shall be connected, either directly or indirectly, to form an equipotential bonding network of such conductors connected to that main earthing terminal/connection or bar:

- (a) **Protective earthing conductors.**

TABLE 5.2
ACCEPTABLE EARTH ELECTRODES

Material	Surface treatment	Minimum dimensions	Minimum surface treatment thickness
Vertical electrodes			
Steel	Copper clad	Ø12 mm circular rod	250 µm
	Copper plated	Ø12 mm circular rod	250 µm
	Stainless (including clad with stainless)	Ø12 mm circular rod	500 µm
	Hot dipped galvanized	Ø16 mm circular rod	63 µm
	Hot dipped galvanized	Section with minimum cross-sectional area of 200 mm ² and with no part less than 3 mm thick	63 µm
Non-ferrous (excluding aluminium)	Solid	12 mm	N/A
Horizontal (strip) electrodes			
Copper rod	Solid	Ø7 mm circular	N/A
Copper strip	Solid	25 mm × 1.6 mm	N/A
Copper pipe	Bare	Ø15 mm circular × 2.45 mm wall thickness	N/A
Copper cable	Bare	25 mm ²	N/A
Steel pipe	Hot dipped galvanized	Ø20 mm circular × 3 mm wall thickness	63 µm
Steel strip	Hot dipped galvanized	40 mm × 3 mm	63 µm

5.3.6.3 Installation

Vertical-type earth electrodes shall be driven to a minimum depth of—

- (a) in Australia, 1.2 m; and
- (b) in New Zealand, 1.8 m.

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Strip-type earth electrodes buried in a horizontal trench shall be laid at a depth having not less than 0.5 m cover and with a minimum horizontal length of—

- (i) in Australia, 3 m; and
- (ii) in New Zealand, 7.5 m.

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(iii) supplied—

- (A) from a source located outside Zone 0;
- (B) at a nominal voltage not exceeding 12 V a.c. or 30 V ripple-free d.c.; and
- (C) as an SELV or a PELV system, in accordance with Clause 7.5.

- (b) *Zones 1, 2 and 3* Appliances and other electrical equipment installed in Zones 1, 2 and 3 shall be provided with at least the required degree of protection for the particular zone.

Heating cable systems intended for heating the location, and that are embedded in the floor and protected in accordance with Clause 4.10, may be installed.

6.2.4.6 Switchboards

A switchboard shall not be located within any classified zone.

6.2.4.7 Electricity generation systems

- * Electricity generation systems, including engine-driven generator sets, stand-alone power systems, grid-connected inverter systems and battery systems shall not be installed within any classified zone.

TABLE 6.1

**SELECTION AND INSTALLATION OF ELECTRICAL EQUIPMENT
FOR BATHS, SHOWERS AND OTHER FIXED WATER CONTAINERS**

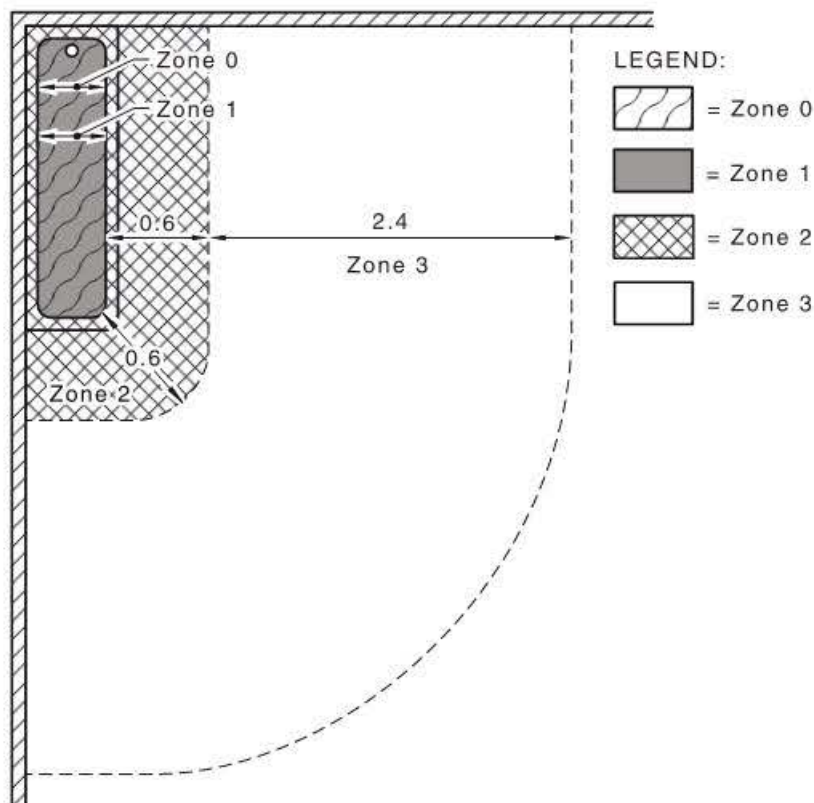
Equipment	Zone 0	Zone 1	Zone 2	Zone 3
Clause 6.2.4.2— Socket-outlets	Not permitted	Not permitted	(a) A shaver outlet; or (b) RCD-protected and in a cupboard (no specific IP rating).	(a) <0.3 m not permitted (b) ≥0.3 m no IP rating† but shall have— (i) RCD protection; (ii) separated supply; or (iii) SELV or PELV supply.
Clause 6.2.4.3— Switches/ accessories	Not permitted	Same as Zone 2	<0.3 m not permitted ≥0.3 m IPX4†	<0.3 m not permitted ≥0.3 m—No IP rating†

(continued)

TABLE 6.1 (continued)

Equipment	Zone 0	Zone 1	Zone 2	Zone 3
Clause 6.2.4.4— Luminaires	IPX7 and specifically for use and SELV or PELV supply	IPX4†	IPX4†; or Class II construction (double or reinforced insulation); or SELV or PELV; or recessed into ceiling	No IP rating†
Clause 6.2.4.5— Other electrical equipment	IPX7 and specifically for use and SELV or PELV supply	IPX4†	IPX4†; or recessed into ceiling	No IP rating†
Clause 6.2.4.6— Switchboards	Not permitted	Not permitted	Not permitted	Not permitted
* Clause 6.2.4.7— Electricity generation systems	Not permitted	Not permitted	Not permitted	Not permitted

† Degree of protection IPX5 required in communal baths/showers.



DIMENSIONS IN METRES

FIGURE 6.1 ZONE DIMENSIONS (PLAN)—
BATH WITHOUT SHOWER OR FIXED BARRIER

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6.3.4.7 *Electricity generation systems*

- * Electricity generation systems, including engine-driven generator sets, stand-alone power systems, grid-connected inverter systems and battery systems shall not be installed within any classified zone.

6.3.4.8 *Electricity distributor's electrical equipment*

- * Pools and spas shall not be located in areas containing electrical equipment, including connection pits and cabinets, owned by the electricity distributor, that result in such equipment being incorporated into any classified zone.

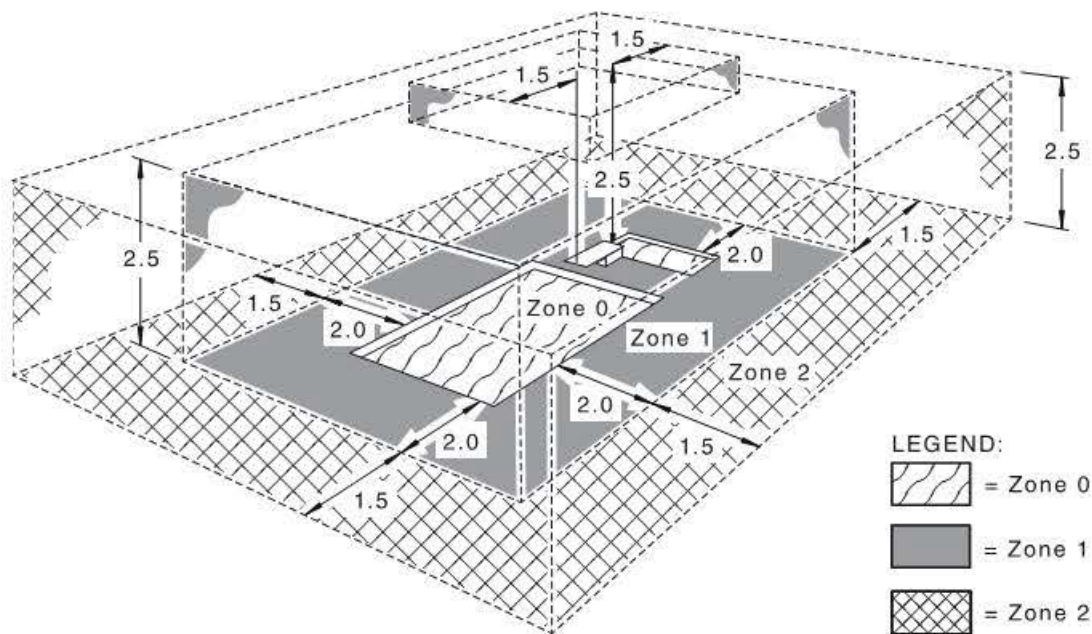
TABLE 6.2
SELECTION AND INSTALLATION OF ELECTRICAL EQUIPMENT
FOR SWIMMING POOLS, PADDLING POOLS AND SPA POOLS
OR TUBS

Equipment	Zone 0	Zone 1	Zone 2
Clause 6.3.4.3— Socket-outlets	Not permitted	Not permitted for general use For pool equipment only: (a) IPX5. (b) (i) ≥ 0.45 m high; and ≥ 1.25 m from internal rim; or (ii) under and ≥ 0.5 m from edge of fixed continuous horizontal barrier ≥ 1.25 m wide. (c) (i) separated supply; (ii) SELV or PELV supply; or (iii) RCD protection.	IPX4 and— (a) separated supply; (b) SELV or PELV supply; or (c) RCD protection.
Clause 6.3.4.4— Switches/accessories	Not permitted	IPX5	IPX4
Clause 6.3.4.5(a), (b) and (c)—Luminaires, appliances and other equipment	IPX8; and specifically for use; and 12 V a.c./30 V d.c. SELV or PELV supply from source outside zone No earth	IPX5 and— (a) SELV or PELV supply; (b) Class II construction (double or reinforced insulation); or (c) Class I construction (earthed conductive parts), fixed in position and RCD protection.	IPX4 and— (a) SELV or PELV supply; (b) separated supply; (c) Class II construction (double or reinforced insulation); or (d) Class I construction (earthed conductive parts), and RCD protection.
Clause 6.3.4.5(d)— Heating cable systems	Not applicable	Permitted where embedded in the floor area under the zone and protected to Clause 4.10.5	As for Zone 1

(continued)

TABLE 6.2 (continued)

Equipment	Zone 0	Zone 1	Zone 2
Clause 6.3.4.6— Switchboards	Not permitted	Not permitted	Not permitted
* Clause 6.3.4.7— Electricity generation systems	Not permitted	Not permitted	Not permitted
* Clause 6.3.4.8— Electricity distributor's electrical equipment	Not permitted	Not permitted	Not permitted



DIMENSIONS IN METRES

FIGURE 6.15 ZONE DIMENSIONS OF IN-GROUND SWIMMING POOLS